



External API v1.0.2

EHR Integration Documentation for secure access to NHS GP records

ORB API v1

ORB External API – Integration guide for EHR systems

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Overview

ORB has been designed to support integration with other clinical systems, usually EHR systems.

In the typical case, clinicians in patient-facing organisations such as Clinics or Sporting Clubs will be using an EHR system to hold patient records, and they would like access to NHS records on those patients. Integrating ORB allows this to happen within the EHR.

This document defines the **ORB External API** for EHR integrations. The API has two main parts:

- Data Model Linkage - the API connects Organisations, Users and Patients from the EHR into ORB
- Data Access - the API allows the EHR system to show specific Patient data when requested by a specific User

Data Model

The API and ORB work with the following data model:

- Organisations - companies using the EHR which own/run clinics
- Users - People in the EHR linked to ORB via the integration (clinical and non-clinical)
- Patients - Individuals treated by the Clinicians

Note that an Organisation is usually not a clinic - it's the company that owns or runs clinics.

To integrate with ORB, EHR developers must use the API to create ORB data records corresponding to each Organisation, User and Patient - as this is necessary for ORB to work. The Organisation and User records are lightweight - they contain only the data sent by the EHR, and they aren't made available in the ORB UI. ORB will not update data held on these records, so there is no need to sync back from Orb, but we do have a ListUsers API to retrieve current users.

Patient records are used to invite the patient to register and share NHS data with the Organisations, and this data is accessible to the patient in the UI.

This API is **organisation-scoped**: all patient and user operations occur in the context of an Organisation.

This API uses external EHR IDs for Organisations, Users and Patients - internally there is a corresponding ORB ID but these are not published. This is to avoid the need for EHRs to store an additional ID.

The API assumes that EHR IDs always refer to the same data item, but otherwise tries not to make assumptions. In particular, User records are assumed to be independent per Organisation - if a single user works at two organisations, this user has two records in ORB which are not linked.

Data Access

The second part of the API concerns access to Patient Data. The ORB API includes a JS Widget that the EHR developer should add to their Patient details page.

When a user wishes to open the widget to view patient data, the EHR must make a server-side request to obtain a short-lived access token linked to that user and that patient. This is then passed to the widget.

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Authentication

All API communication must use **HTTPS** (TLS 1.2 or higher recommended). Do not send API keys or patient data over unencrypted connections.

All requests must include the following header:

```
X-API-KEY: <your-api-key>
```

Base URL

The ORB API is available at the following base URL:

- <https://api-test-app.orbforhealth.com/v1>

Access to base URLs will be provided when your API key is issued.

Route prefixes (implementation):

- **Organisation** and **User** endpoints: [v1/organisations/{extOrganisationId}](#)
- **Patient** endpoints: [v1/organisations/{extOrganisationId}/patients](#)

For example, to create an organisation in the test environment, you would make a POST request to:

```
https:// https://api-test-app.orbforhealth.com/v1/organisations/{extOrganisationId}
```

API Keys

- API Keys are issued by ORB
 - Each API Key is linked to **ONLY one External EHR**
 - Within ORB this is called a Governing Body (GovBody) A
 - GovBody may contain multiple Organisations
 - API Keys **cannot** be linked directly to Organisations
-

Conventions

- All requests and responses use JSON; requests with a body should use **Content-Type: application/json**
- Dates are ISO-8601 formatted
- External IDs (**extOrganisationId**, **extUserId**, **extPatientId**) are provided by the EHR and treated as stable
- ORB does **not** expose internal ORB IDs - so EHR does not need to store them
- Deleted records are retained for legal purposes, but are not re-used and are inaccessible

Key terms

Term	Meaning
GovBody	Governing Body; the EHR (or partner) linked to one API key. A GovBody contains many Organisations.
External ID	extOrganisationId , extUserId , or extPatientId — the identifier from the EHR. ORB does not publish its internal IDs.
ConnectionStatus	Patient's NHS login/connection state (e.g. InviteNotSent, InviteSent, Connected, DataMismatch). See Patients section.
DataAccessStatus	Whether the patient has been asked to grant access and whether they have granted (RequestNotSent, RequestSent, Reviewed).

Common Errors

The following HTTP status codes may be returned by any endpoint in addition to endpoint-specific errors:

Status	Meaning
500	Internal server error - an unexpected error occurred on the server
Status	Meaning
503	Service unavailable - the service is temporarily unavailable, please retry later

For 503, use exponential backoff when retrying.

For other 4xx and 5xx responses, the body may include a JSON object describing the error.

For **400 Bad Request** when the request cannot be parsed or accepted—for example malformed JSON, a required body field missing, a query or path value in the wrong form (such as `offset=0sd` when an integer is expected), a value outside the permitted set (such as `"sex": "Female1"` when only defined values are accepted), or a property not defined for that operation (such as an unknown field `"newField"` in a **PATCH** or **POST** request body)—and no more specific ORB **errorCode** applies, the response always includes **errorCode** `InvalidRequestData` and a single **errorMessage**. The response contains **only** these two members.

errorMessage is intended for display. It is either a fixed general sentence or—commonly—a short line naming one query parameter and the rejected value when only that parameter is at fault (see **InvalidRequestData** below). Wording may change in later API versions; use **errorCode** for programmatic handling and **do not** parse **errorMessage** for business logic.

Property	Type	Description
errorMessage	string	Human-readable error message. Example: "Organisation not found".
errorCode	string?	Optional. Present when the exception is a known ORB API exception; see values below.

Example response body:

```
{
  "errorMessage": "Organisation not found",
  "errorCode": "OrganizationNotFound"
}
```

Possible **errorCode** values:

- ♦ **ClientNotFound** - The caller client is not registered in ORB or the corresponding governing body record is missing for some reason. Cannot validate API key nor allow further operations from this caller.
- ♦ **OrgNotFound** - There is no existing organisation corresponding to the provided id. **OrgLinkNotFound** - User or patient is not linked to the organization corresponding to the provided id. **UserNotFound** - There is no existing user corresponding to the provided id.
- ♦ **PatientNotFound** - There is no existing patient corresponding to the provided id.
- ♦ **InvalidRequestData** — Returned when the request body, query string, or path cannot be parsed or validated successfully, and no more specific ORB **errorCode** applies. The response body is always the compact JSON object { "errorCode", "errorMessage" } described under **Common Errors** above. **errorMessage** is assigned as follows:

- ◊ **General message** — Used when more than one validation issue is reported for the same request, and in typical cases where the problem lies in the **JSON request body** only—including a value outside a permitted enumeration (for example "sex": "Female1" when only defined values are accepted). In these cases **errorMessage** is a single standard sentence; clients must not assume a field-by-field breakdown in the message text.

```
{
  "errorCode": "InvalidRequestData",
  "errorMessage": "One or more fields contain invalid values. Please
verify the input data."
}
```

- ◊ **Per-parameter message (query)** — Typically used when a **single** query (or equivalent) parameter does not match its expected type; **errorMessage** may then identify the parameter and the supplied value:

```
{
  "errorCode": "InvalidRequestData",
  "errorMessage": "The value '0sd' is not valid for Offset."
}
```

Other isolated query-parameter errors may follow the same pattern. The exact wording of **errorMessage** is not guaranteed across API versions; use it for display only and base integration logic on **errorCode**.

- **InvalidProperty** - Some request body field holds an invalid value (used when the API returns a specific semantic validation outcome rather than the generic **InvalidRequestData** body above).
- **InvalidParameter** - Some request parameter holds an invalid value.
- **IdNotAvailable** - The id provided is already in use by another entity of the same type. For example, client tried to create a new organization using the same id of an existing one.
- **NameNotAvailable** - The name provided is already in use by another entity of the same type. For example, client tried to create a new organization using the same name of an existing one.
- **EmailNotAvailable** - The email provided is already in use by another entity of the same type. For example, client tried to create a new user using the same email of an existing one.
- **DobMismatch** - There is a DOB mismatch between the patient data in the request and the data of the corresponding patient in ORB.
- **SurnameMismatch** - There is a surname mismatch between the patient data in the request and the data of the corresponding patient in ORB.
- **PatientInvalidStatus** - The operation requested cannot be performed given the current status of the patient.
- **ResendLimitExceeded** - The client has exceeded the maximum number of requests allowed in a given period of time.
- **NonClinicianAccess** - A patient access token was requested for a user with `isClinician: false` (only clinical users may request access tokens).
- **OrganisationContractNotSigned** - the organisation has not completed ORB organisation contract signing. Protected EHR API and widget operations for that organisation remain blocked until the contract is signed.
- **GenericError** - The current error does not contain relevant information for the client. Often, this is just an internal logic error of the server, and does not require actions on the client side.

For unexpected or non-API exceptions, **errorCode** may be omitted. Clients can use **errorMessage** for display and optionally use **errorCode** for programmatic handling.

Organisations

Each EHR organisation must complete the before most Users, Patients, and Widget operations are available.

Create Organisation

`POST /organisations/{extOrganisationId}`

Creates an Organisation in ORB under the GovBody associated with the API Key. Creation is allowed before the organisation contract is signed (required for onboarding).

On create, ORB sets default NHS linking welcome text used on the patient-facing EHR linking page (see Set Organisation Branding):

- Welcome title: Welcome to ORB
- Welcome subtitle: Please login/register with your NHS account to continue.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)

Request Body

```
{
  "organisationName": "string",
  "address1": "string",
  "address2": "string",
  "address3": "string",
  "address4": "string",
  "postcode": "string",
  "phoneNumber": "string",
  "cqcRegistrationNumber": "string",
  "researchOptOut": boolean
}
```

Request Body Fields

Field	Type	Required	Description
organisationName	string	Yes	Unique Organisation name (case-insensitive)
address1	string	Yes	First line of address
address2	string	Yes	Second line of address
address3	string	No	Third line of address
address4	string	No	Fourth line of address
postcode	string	Yes	Postcode
phoneNumber	string	Yes	Phone number
cqcRegistrationNumber	string	Yes	CQC registration number
researchOptOut	boolean	No	Research opt-out flag (defaults to false/off)

Notes:

- ext OrganisationId must be unique within the GovBody.
- organisationName must be unique within the GovBody (case-insensitive).

Responses

Status	Meaning
201	Organisation created
400	Creation failed due to invalid request or validation error
404	No external client found for the provided API key
409	Organisation already exists with the same extOrganisationId or organisationName

Update Organisation

PATCH /organisations/{extOrganisationId}

Updates organisation details or suspends / unsuspends the organisation. Requires the organisation contract to be signed.

- ♦ If an organisation is suspended it is not considered for billing, and no Users can access data via API. Patients are unaffected - Patients can still login/register with the Organisation via the ORB UI. Organisations cannot be
- ♦ suspended indefinitely. When suspending, the caller may specify **autoDeleteDays** to set when automatic deletion happens. If not specified this happens in 180 days. Organisations can be unsuspended using this
- ♦ feature to restore access for Users accessing data via API, and will again be considered for billing.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)

Request Body

```
{
  "organisationName": "string",
  "address1": "string",
  "address2": "string",
  "address3": "string",
  "address4": "string",
  "postcode": "string",
  "phoneNumber": "string",
  "cqcRegistrationNumber": "string",
  "researchOptOut": boolean,
  "suspended": boolean,
  "autoDeleteDays": number
}
```

Request Body Fields

Field	Type	Required	Description
organisationName	string	No	Unique Organisation name (case-insensitive)
address1	string	No	First line of address
address2	string	No	Second line of address
address3	string	No	Third line of address
address4	string	No	Fourth line of address
postcode	string	No	Postcode
phoneNumber	string	No	Phone number
cqcRegistrationNumber	string	No	CQC registration number
researchOptOut	boolean	No	Research opt-out flag
suspended	boolean	No	Suspended flag
autoDeleteDays	number	No	Auto-delete after suspension in days (max 180, default 180)

Responses

Status	Meaning
200	Organisation updated
400	Update failed due to invalid request or validation error
403	Organisation contract not signed ('OrganisationContractNotSigned')
404	Organisation corresponding to given id or Governing Body corresponding to given API key not found
409	Organisation Name already exists

Delete Organisation

DELETE /organisations/{extOrganisationId}

Delete the organisation and all associated users. Requires the organisation contract to be signed. This should be used if an organisation asks to turn OFF the integration permanently or if it leaves the EHR system.

Once deleted:

- The extOrganisationId is no longer valid for use in any API calls.
- If a patient has no other organisations linked to their profile, they will no longer be able to login.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)

Responses

Status	Meaning
200	Organisation deleted
400	Deletion failed due to invalid request or validation error
403	Organisation contract not signed ('OrganisationContractNotSigned')
404	Organisation not found

A successful response (200) returns no response body.

Get Organisation

GET /organisations/{extOrganisationId}

Returns the organisation details for the given external organisation ID when it exists in the Governing Body linked to the API key.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)

Response Body

A successful response (200) returns the organisation summary:

```
{
  "extOrganisationId": "string",
  "organisationName": "string",
  "suspended": false,
  "autoDeleteDate": "string"
}
```

Response Body Fields

Field	Type	Description
extOrganisationId	string	External organisation ID (EHR)
organisationName	string	Organisation name
suspended	boolean	Suspension status
autoDeleteDate	string?	Auto-delete date in YYYY-MM-DD format when suspended; null if not set

Note: autoDeleteDate is only populated when suspended = true.

Responses

Status	Meaning
200	Organisation exists in Governing Body
400	Bad request
404	Organisation not found

List Organisations

GET /organisations

Returns a list of organisations under the Governing Body linked to the API key. Includes suspension status and auto-delete date when suspended. Filtering by suspension status is optional.

Query Parameters

Name	Type	Required	Default	Description
suspended	boolean	No	-	Filter by suspension status. Omit to return all results

Response Body

```
[
  {
    "extOrganisationId": "string",
    "organisationName": "string",
    "suspended": boolean,
    "autoDeleteDate": "string"
  }
]
```

Response Body Fields

Field	Type	Description
[].extOrganisationId	string	External organisation ID (EHR)
[].organisationName	string	Organisation name
[].suspended	boolean	Suspension status
[].autoDeleteDate	string	Auto-delete date in YYYY-MM-DD format when suspended; null if not set

Note: autoDeleteDate is only populated when suspended = true.

Example Requests

Get all organisations:

```
GET /organisations
```

Get only suspended organisations:

```
GET /organisations?suspended=true
```

Get only active (not suspended) organisations:

```
GET /organisations?suspended=false
```

Responses

Status	Meaning
200	OK
404	Governing Body corresponding to given API key not found

Get Organisation Contract Status

[GET /organisations/{extOrganisationId}/contract-status](#)

Returns whether the organisation has signed the ORB organisation contract and when. The host uses this to decide whether to show the contract signing iframe and to verify status after signing. Available before the contract is signed (not gated).

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)

Response Body

```
When not signed:
{
  "extOrganisationId": "org-
123",
  "contractSigned": false,
  "contractSignedAt": null
}
When signed:
{
  "extOrganisationId": "org-
123",
  "contractSigned": true,
  "contractSignedAt": "2026-05-
22T10:30:00Z"
}
```


Response Body Fields

Field	Type	Description
extOrganisationId	string	Echo of path parameter
contractSigned	boolean	true if the organisation contract has been signed
contractSignedAt	string null	ISO-8601 timestamp of signing; null if not signed

Responses

Status	Meaning
200	OK
401	Missing or invalid API key
403	EHR API disabled for this client
404	Organisation not found for this API key / client
500	Unexpected server error

Request Contract Access Token

POST /organisations/{extOrganisationId}/contract-access-token

Mints a short-lived organisation-scoped token used to open the ORB organisation-contract signing iframe. Available before the contract is signed (not gated). The token identifies the API-key client and external organisation only (no patient or user claim).

The host must use a valid API key and the same extOrganisationId that was passed when the organisation was created. ORB validates that the organisation belongs to that client before issuing the token.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)

Request Body

None.

Response Body

Response Body Fields

Field	Type	Description
accessToken	string	JWT for the contract-signing iframe session
expiresIn	integer	Token lifetime in seconds

Responses

```
{
  "accessToken": "string",
  "expiresIn": 3600
}
```

Status	Meaning
200	Token issued
401	Missing or invalid API key
403	Organisation suspended, or EHR API disabled for this client
404	Organisation not found for this API key / client
500	Unexpected server error

Embedding, postMessage, and the in-iframe signing UI are documented under Organisation contract (clinic) and in the ORB Widget Integration Guide.

Set Organisation Branding

POST /organisations/{extOrganisationId}/branding

Sets organisation branding used on the patient-facing NHS linking page opened from Connect Patient by Email / Connect Patient by Link: organisation logo and optional welcome title/subtitle.

Requires the organisation contract to be signed (403 OrganisationContractNotSigned otherwise).

Request is multipart/form-data (not JSON). A successful call returns 204 with no body.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)

Request Body

Send as multipart/form-data (this endpoint accepts a file upload, so JSON body is not used).

Field	Type	Required	Description
overwriteLogo	boolean	No	When true, update the stored logo: send a file to replace it, or send no file to remove the current logo. When false or omitted, the existing logo is left unchanged (any uploaded file is ignored).
file	binary	No	Logo image. Allowed: PNG, JPEG, JPG, SVG. Maximum size: 1 MB. Only one logo is stored; uploading another replaces the previous.
patientNHSActivationWelcomeTitle	string	No	Welcome title on the NHS linking page. Updated only when this field is included.
patientNHSActivationWelcomeSubtitle	string	No	Welcome subtitle on the NHS linking page. Updated only when this field is included.

You may update welcome text without changing the logo (omit overwriteLogo, or set it to false).

You may change the logo without sending welcome fields.

Defaults and patient experience

If welcome text has not been customised, patients see the create-organisation defaults (Welcome to ORB / Please login/register with your NHS account to continue.).

On the linking page the patient sees:

- the organisation logo when one is uploaded, otherwise ORB's default branding mark;
- the welcome title and subtitle;
- an **NHS Login** button only with the self register option

Responses

Status	Meaning
204	Branding updated (no response body)
400	Invalid request or validation error
403	Organisation contract not signed (OrganisationContractNotSigned), or EHR API disabled
404	Organisation or Governing Body not found
415	Unsupported media type (file not PNG/JPEG/JPG/SVG, content mismatch, or not multipart)

Users

Users represent people in the EHR who are linked to ORB via the integration. They do not have general login access to the ORB UI.

Required behaviour (isClinician)

- isClinician is mandatory when creating a user via the API.
- isClinician is optional when updating a user.
- isClinician is always returned in user list responses.

Each user record includes three boolean flags:

- **isClinician** — `true` if the user is a clinical user who may access NHS patient data via the integration (widget / access token); `false` for non-clinical users (for example administrative staff).
- **isLocum** — `true` if the user is a temporary locum; `false` if a permanent employee.
- **authorisedSignatory** — `true` if this user is an authorised signatory for the organisational contract.

All three flags are required when creating a user; any of them may be supplied on update.

User create, update, list, and delete require the organisation contract to be signed («403» **OrganisationContractNotSigned** otherwise).

Create User

POST /organisations/{extOrganisationId}/users/{extUserId}

Creates an ORB user within an ORB organisation, linked to the corresponding external User ID.

This is used to grant a user access to the integration. Set isClinician to `true` for clinical users and `false` for non-clinical users. Only users with isClinician: true may request patient access tokens (see [Request Access Token](#request-access-token)). For Users we assume the external IDs are unique within an Organisation, and do not change.

- We do not attempt to link or merge users – nor do we identify duplicates based on personal attributes (e.g. FirstName, LastName, ProfessionalRegNumber).
- If the same clinician exists in multiple organisations in the EHR, a separate ORB user must be created for each organisation, even if in EHR they are shared.

The number of Users in existence during a month may be used in billing. The system logs user creation and deletion and this log is available for billing.

Path Parameters

Name	Description
extOrganisationId	External Organisation ID (EHR)
extUserId	External User ID (EHR)

Request Body

```
{  
  "firstName": "string",  
  "lastName": "string",  
  "professionalRegNumber": "string",  
  "emailAddress": "string",  
  "isClinician": true,  
  "isLocum": false,  
  "authorisedSignatory": false  
}
```

Request Body Fields

Field	Type	Required	Description
firstName	string	Yes	User's first name
lastName	string	Yes	User's last name
professionalRegNumber	string	Yes	Professional registration number
emailAddress	string	No	User's email address
isClinician	boolean	Yes	`true` if the user is a clinical user; `false` for non-clinical users (e.g. administrative)
isLocum	boolean	Yes	if the user is a temporary locum; false if a permanent employee
authorisedSignatory	boolean	Yes	true if this user is an authorised signatory for the organisational contract

Responses

Status	Meaning
201	User created
400	Creation failed due to invalid request or validation error
404	Organisation not found
409	User already exists in this organisation with the same extUserId

Update User

PATCH /organisations/{extOrganisationId}/users/{extUserId} Updates one or

more attributes of an existing user within an organisation. Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)
extUserId	External user ID (EHR)

Request Body

```
{
  "firstName": "string",
  "lastName": "string",
  "professionalRegNumber": "string",
  "emailAddress": "string",
  "isClinician": true,
  "isLocum": false,
  "authorisedSignatory": false
}
```

Responses

Field	Type	Required	Description
firstName	string	No	User's first name
lastName	string	No	User's last name
professionalRegNumber	string	No	Professional registration number
emailAddress	string	No	User's email address
lastName	string	No	User's last name
isClinician	boolean	Yes	`true` if the user is a clinical user; `false` for non-clinical users (e.g. administrative)
isLocum	boolean	No	If supplied, updates whether the user is a locum (true)
authorisedSignatory	boolean	No	If supplied, updates authorised signatory flag for the organisational contract

Status	Meaning
200	User updated
400	Invalid request or validation error
404	User or organisation not found

List Users

GET /organisations/{extOrganisationId}/users

Returns users for the organisation (regardless of whether the organisation is currently suspended). Pagination is optional - if no pagination parameters are provided, all users are returned as an array. If pagination parameters are provided, a paginated response with metadata is returned.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)

Query Parameters

Name	Type	Required	Default	Description
limit	integer	No	-	Maximum number of users to return (1-1000). If omitted, all users are returned
offset	integer	No	0	Number of users to skip before returning results. Only used when limit is provided

Response Body (No Pagination)

When no pagination parameters are provided, the response is a simple array:

```
[
  {
    "extUserId": "string",
    "firstName": "string",
    "lastName": "string",
    "professionalRegNumber": "string",
    "emailAddress": "string",
    "isClinician": true,
    "isLocum": false,
    "authorisedSignatory": false
  }
]
```

Response Body (With Pagination)

When **limit** is provided, the response includes pagination metadata:

```
{
  "users": [
    {
      "extUserId": "string",
      "firstName": "string",
      "lastName": "string",
      "professionalRegNumber": "string",
      "emailAddress": "string",
      "isClinician": true,
      "isLocum": false,
      "authorisedSignatory": false
    }
  ],
  "pagination": {
    "total": 150,
    "limit": 100,
    "offset": 0,
    "hasMore": true
  }
}
```

Response Body Fields

When pagination is not used (array response):

Field	Type	Description
[].extUserId	string	External user ID (EHR)
[].firstName	string	User's first name
[].lastName	string	User's last name
[].professionalRegNumber	string	Professional registration number
[].emailAddress	string	User's email address
users[].isLocum	boolean	true if locum; false if permanent employee
[].isLocum	boolean	true if locum; false if permanent employee
[].authorisedSignatory	boolean	true if authorised signatory for the organisational contract

When pagination is used (object response):

Field	Type	Description
users	array	Array of user objects
users[].extUserId	string	External user ID (EHR)
users[].firstName	string	User's first name
users[].lastName	string	User's last name
users[].professionalRegNumber	string	Professional registration number
users[].emailAddress	string	User's email address
users[].isClinician	boolean	`true` if clinical user; `false` if non-clinical
users[].isLocum	boolean	true if locum; false if permanent employee
users[].authorisedSignatory	boolean	true if authorised signatory for the organisational contract
pagination	object	Pagination metadata
pagination.total	integer	Total number of users in the organisation
pagination.limit	integer	Number of users requested per page
pagination.offset	integer	Number of users skipped
pagination.hasMore	boolean	Whether there are more users available

Example Requests

Get all users (no pagination):

```
GET /organisations/ORG123/users
```

Get first 100 users (with pagination):

```
GET /organisations/ORG123/users?limit=100&offset=0
```

Get next 50 users:

```
GET /organisations/ORG123/users?limit=50&offset=100
```

Responses

Status	Meaning
200	OK
400	Invalid pagination parameters (limit > 1000, limit < 1, or offset < 0)
404	Organisation not found

Delete User

DELETE /organisations/{extOrganisationId}/users/{extUserId}

Deletes the specified user from the organisation. After deletion, the user will no longer be able to access the integration for this organisation.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)
extUserId	External user ID (EHR)

Responses

Status	Meaning
200	User deleted
404	User or organisation not found

A successful response (200) returns no response body.

Patients

Patients are linked to organisations via a **connection** resource. Patient list, update, connection, permissions, and access-token operations for an organisation require the organisation contract to be signed (403 **OrganisationContractNotSigned** otherwise).

ORB registers a couple of statuses and some flags for each patient, to be able to provide relevant information about the patient connection process to the external client:

- ♦ **ConnectionStatus**: Describes the process of requesting NHS registration of the patient from the external organisation. The valid values are:
 - ◊ **InviteNotSent**: Invitation for the patient to log in using NHS credentials is not sent yet. Valid only if the patient has never logged in using NHS.
 - ◊ **InviteSent**: Invitation for the patient to log in with their NHS credentials has been sent. Valid only if the patient has never logged in with the NHS.
 - ◊ **InviteExpired**: Invitation for the patient to log in with their NHS credentials has been sent but expired. Valid only if the patient has never logged in with the NHS.
 - ◊ **DataMismatch**: Patient tried to log in with their NHS credentials, but there was a mismatch between the data they have in ORB and the data they have in NHS. If this status is set, two other fields provide the client with information about which field contains the mismatch.
 - **DOBMismatch**: This field is true if there is a mismatch in the date of birth.
 - **SurnameMismatch**: This field is true if there is a mismatch in the surname.
 - ◊ **Connected**: Patient did a valid NHS registration, and there are no mismatches. This only indicates patient has NHS login and there is an external patient linked to it. It has nothing to do with permissions to access patient NHS data.
- ♦ **DataAccessStatus**: Describes the process of requesting access to the NHS data of the patient from the external organisation. Their valid values are:
 - ◊ **RequestNotSent**: Request for accessing patient NHS data is not sent yet.
 - ◊ **RequestSent**: Request for accessing patient NHS data has been sent.
 - ◊ **Reviewed**: The patient granted the organisation access to his NHS data. This only indicates the patient allowed access to **some** of their NHS data. To know exactly to which data the organisation has access, client must call **Get Granted Permissions** API method. Also, patient data can be blocked from GP side or by the patient. If this status is set, two other fields provide this info.
 - **PatientDataAccessLimited**: This field is true if the patient has restricted access to their data partially or totally (e.g. via passport settings).
 - **GpDataAccessLimited**: This field is true if the GP blocked the patient's data partially or totally in any way.
- ♦ **Oldest NHS Record**: The date of the oldest record received from NHS for this patient. If this is relatively recent (e.g. one year ago), it may indicate that the GP has restricted how far back access goes. This is returned as **oldestNhsRecordDate** in API responses; it may be null if no data has been received yet or cannot be determined.

Data priority and overwrite rules (ORB vs EHR data)

A patient may have data from NHS, from ORB (entered by other organisations), and from the EHR/organisation via the API (name, surname, DOB, postcode, etc.). ORB applies the following rules for all patients, including those already existing in ORB:

- ♦ **NHS data** has the highest priority and is always considered most valid.
- ♦ **EHR/organisation data** (sent via the API when linking or connecting a patient) has higher priority than ORB-held data for fields where NHS has not provided a value.

When the EHR uses the API to link or connect a patient and supplies data in the request body:

- ◆ **Date of birth and surname** (`dateOfBirth`, `lastName`): These are **validated** against existing ORB data, not overwritten. If the values differ from what is already stored (ORB vs request mismatch), the request fails with **409** (conflict). This applies when the patient already exists in ORB (e.g. matched by email). This is distinct from the **NHS→ORB mismatch** (when the patient has tried NHS login and ORB data did not match NHS), which is reported as **422** with a response body containing `connectionStatus` **DataMismatch** (see Connect Patient by Email / by Link and Get Patient Connection Status).
- ◆ **All other supplied fields** overwrite ORB data unless NHS has already provided a value for that field (NHS takes precedence). The fields that can be overwritten when supplied by the EHR on Connect

Patient by Email or Connect Patient by Link are:

- `title`
- `firstName`
- `sex`
- `emailAddress`
- `postcode`

Example: if the EHR supplies a postcode and NHS has not provided one, the EHR postcode overwrites any existing ORB postcode for that patient.

ORB uses the following process when a connection request is made, updating patient statuses each time:

1. First, it checks if the patient already exists in the system, using the email as key. ORB does not support two patients with the same email address; there is at most one patient per email.
 - If the patient does not exist, ORB creates the patient record and the external link record for this organisation, then continues.
 - If the patient exists (email match), it checks if the date of birth and surname sent in the request match the ones of the existing patient (see Data priority above).
 - If they don't match, an error is returned to the EHR and the request fails (409). Surname and DOB are not overwritten in this case.
 - Otherwise, any other supplied fields overwrite ORB data according to the data priority rules above.
 - Patient connection status is changed to **InviteNotSent**.
2. Then, it checks if the patient has NHS login.
 - If not, sends an ORB NHS registration email to the user, and waits for the user to click the link and sign up using NHS Login.
 - If the email has been sent before (patient is in **InviteSent** status), and it has not expired yet, a `resend` flag must be included in the body of the request to indicate that client wants to resend the email anyway.
 - This email can be sent at most twice per 24 hours, to avoid spamming the patient with too many of these emails.
 - Patient connection status is changed to **InviteSent**.

3. When the patient tries to log in using NHS, ORB compares the surname and DOB held in ORB with the NHS details.
 - If they don't match: ORB overwrites the patient record with the correct surname and DOB from NHS (so the patient can complete NHS login). The **external link** for the organisation that registered this patient is then set to **DataMismatch**, recording which field(s) failed (**dobMismatch**, **surnameMismatch**). So the mismatch is recorded per organisation; the master patient data in ORB is corrected from NHS.
 - If a second organisation later registers the same patient (same email), ORB data now matches NHS, so the request can succeed.
 - If they match, patient connection status is set to **Connected** (step 4).
4. If there are no mismatches, patient connection status is changed to **Connected**.
5. Then, it asks the patient to grant NHS access permissions to the requesting Org.
 - An email is sent for this, which redirects to patient passport access page after login. After this, patient data access request status is changed to **RequestSent**.
 - If the email has been sent before (patient is in **RequestSent** status), a **resend** flag must be included in the body of the request to indicate that client wants to resend the email anyway. This email can be sent at
 - most twice per 24 hours, to avoid spamming the patient with too many of these emails.
 - If patient allows any access to their data, patient data access request status is changed to **Reviewed**.

Note that patient can allow data access partially, and their data can still be blocked from GP.

Rate limits (emails and links)

- ◆ **Data access request:** At most two data access request emails (or links) per patient per organisation per 24 hours. If the EHR calls again after that limit has been reached within the period, no new data access email is sent and no data access link is returned.
- ◆ **NHS invitation:** At most two NHS invitation emails (or links) per patient per organisation per 24 hours. If the EHR calls again after that limit has been reached within the period, no new invitation email is sent and no invitation link is returned.

Link validity and renewal

The two links returned or sent by the connection endpoints have different validity rules:

Link type	Used when	Validity
NHS registration / invitation link	The patient has not completed NHS registration	The expiry period is system-configurable and is 7 days by default, starting when the link is generated. Email and direct-link flows use the same type of token.
Data-access / consent link	The patient has completed NHS registration and needs to review access for the organisation	The URL has no separate expiry period. It takes the patient to ORB login and requires authentication before access can be reviewed.

Only one current NHS registration token is retained for a patient. When ORB generates a new registration link, the previous registration link is invalidated immediately and the configured expiry period starts again. To request a replacement while an invitation is still pending, set `resend` to `true`. Generation and sending remain subject to the NHS invitation rate limit above.

The `data-access / consent` URL does not contain an expiring invitation token, so requesting it again does not renew an expiry period. Reissuing or resending it is still subject to the data-access request rate limit above.

Guidance for EHR implementers (data access)

Use the following when building your integration:

- ◆ **If the patient has not granted access** (`dataAccessStatus` is not **Reviewed**): You can call the `Connect Patient by Email` or `Connect Patient by Link` endpoint up to twice per 24 hours (per rate limits above); ORB will send (or return a link for) a reminder to the patient while the limit allows. See Rate limits above.
- ◆ **If the patient has granted access** (`dataAccessStatus` is **Reviewed**): You must check the detailed return values to see if you have enough access for your use case. Call **Get Granted Permissions** for the list of data sets the patient has granted. If something you need is missing, trigger a request to the patient to grant more (e.g. via the connection endpoint or your own messaging).
- ◆ **If `oldestNhsRecordDate` is not very old** (e.g. only one or two years back): Consider informing the patient that they may need to ask their GP to allow access to older records, or use in-app messaging to suggest they check GP restrictions.
- ◆ **If `patientDataAccessLimited` is true**: The patient has restricted what data is shared. Consider informing the clinician and the patient may need to grant more access via their passport settings.
- ◆ **If `gpDataAccessLimited` is true**: Warn the clinician. They should check whether the data available is sufficient; if not, they can ask the patient to request that the GP provide more access.
- ◆ **If the granted permissions (`Get Granted Permissions`) do not include what you need**: Tell the clinician which data sets (e.g. allergies, medications) the patient has not granted. Allow the clinician to trigger a request to the patient to grant more if needed.

List Patients

`GET /organisations/{extOrganisationId}/patients`

Returns a list of patients for the specified organisation. Pagination is optional - if no pagination parameters are provided, all patients are returned as an array. If pagination parameters are provided, a paginated response with metadata is returned.

Path Parameters

Name	Description
<code>extOrganisationId</code>	External Organisation ID (EHR)

Query Parameters

Name	Type	Required	Default	Description
limit	integer	No	-	Maximum number of patients to return (1-1000). If omitted, all patients are returned
offset	integer	No	0	Number of patients to skip before returning results. Only used when limit is provided
connectionStatus	string	No	-	Filter by connection status (InviteNotSent, InviteSent, InviteExpired, DataMismatch, Connected)
dataAccessStatus	string	No	-	Filter by data access request status (RequestNotSent, RequestSent, Reviewed)
patientDataAccessLimited	boolean	No	-	When true, filter to patients which have restricted their data (patient-side)
gpDataAccessLimited	boolean	No	-	When true, filter to patients which have their data blocked from GP side
dobMismatch	boolean	No	-	When true, filter to patients whose date of birth in ORB mismatches the one in NHS
surnameMismatch	boolean	No	-	When true, filter to patients whose surname in ORB mismatches the one in NHS

Response Body (No Pagination)

When no pagination parameters are provided, the response is a simple array:

```
[
  {
    "extPatientId": "string",
    "title": "string",

    "firstName": "string",
    "lastName": "string",
    "dateOfBirth": "string",
    "sex": "string",
    "emailAddress": "string",
    "postcode": "string",
    "connectionStatus": "string",
    "dobMismatch": boolean,
    "surnameMismatch": boolean,
    "patientDataAccessLimited": boolean,
    "gpDataAccessLimited": boolean,
    "dataAccessStatus": "string",
    "oldestNhsRecordDate": "string",
    "lastInvitationEmailDate": "string",
    "lastAccessRequestEmailDate": "string"
  }
]
```

Response Body (With Pagination)

When a **limit** is provided, the response includes pagination metadata:

```
{
  "patients": [
    {
      "extPatientId": "string",
      "title": "string",
      "firstName": "string",
      "lastName": "string",
      "dateOfBirth": "string",
      "sex": "string",
      "emailAddress": "string",
      "postcode": "string",
      "connectionStatus": "string",
      "dobMismatch": boolean,
      "surnameMismatch": boolean,
      "patientDataAccessLimited": boolean,
      "gpDataAccessLimited": boolean,
      "dataAccessStatus": "string",
      "oldestNhsRecordDate": "string",
      "lastInvitationEmailDate": "string",
      "lastAccessRequestEmailDate": "string"
    }
  ],
  "pagination": {
    "total": 150,
    "limit": 100,
    "offset": 0,
    "hasMore": true
  }
}
```

Response Body Fields

When pagination is not used (array response):

Field	Type	Description
[].extPatientId	string	External patient ID (EHR)
[].title	string	Patient title
[].firstName	string	Patient's first name
[].lastName	string	Patient's last name
[].dateOfBirth	string	Date of birth in YYYY-MM-DD format
[].sex	string	Patient sex (e.g. Male, Female)
[].emailAddress	string	Patient email address
[].postcode	string	Patient postcode
[].connectionStatus	string	Connection status (InviteNotSent, InviteSent, InviteExpired,
[].dobMismatch	boolean	When connectionStatus is DataMismatch: true if date of birth
[].surnameMismatch	boolean	When connectionStatus is DataMismatch: true if surname
[].patientDataAccessLimited	boolean	When applicable: true if the patient has restricted their data
[].gpDataAccessLimited	boolean	When applicable: true if the GP has blocked the patient's data
[].dataAccessStatus	string	Data access request status (RequestNotSent, RequestSent,
[].oldestNhsRecordDate	string	Date of oldest NHS record received (ISO-8601); null if none or
[].lastInvitationEmailDate	string	ISO-8601 date of the last NHS registration invitation email sent (optional)
[].lastAccessRequestEmailDate	string	ISO-8601 date of the last data access request email sent (optional)

When pagination is used (object response):

Field	Type	Description
patients	array	Array of patient objects
patients[].extPatientId	string	External patient ID (EHR)
patients[].title	string	Patient title
patients[].firstName	string	Patient's first name
patients[].lastName	string	Patient's last name
patients[].dateOfBirth	string	Date of birth in YYYY-MM-DD format
patients[].sex	string	Patient sex (e.g. Male, Female)
patients[].emailAddress	string	Patient email address
patients[].postcode	string	Patient postcode
patients[].connectionStatus	string	Connection status (InviteNotSent, InviteSent,
patients[].dobMismatch	boolean	When connectionStatus is DataMismatch: true if date
patients[].dobMismatch	boolean	When connectionStatus is DataMismatch: true if date
patients[].patientDataAccessLimited	boolean	When applicable: true if the patient has restricted
patients[].gpDataAccessLimited	boolean	When applicable: true if the GP has blocked the
patients[].dataAccessStatus	string	Data access request status (RequestNotSent,
patients[].oldestNhsRecordDate	string	Date of oldest NHS record received (ISO-8601); null if
patients[].lastInvitationEmailDate	string	ISO-8601 date of the last NHS registration invitation (email) sent optional
patients[].lastAccessRequestEmailDate	string	ISO-8601 date of the last data access request email
pagination	object	Pagination metadata
pagination.total	integer	Total number of patients in the organisation
pagination.limit	integer	Number of patients requested per page
pagination.offset	integer	Number of patients skipped
pagination.hasMore	boolean	Whether there are more patients available

Example Requests (combine filters as needed using AND)

Get all patients (no pagination):

```
GET /organisations/ORG123/patients
```

Get only connected patients:

```
GET /organisations/ORG123/patients?connectionStatus=Connected
```

Get connected patients with patient-restricted data access:

```
GET /organisations/ORG123/patients?  
connectionStatus=Connected&patientDataAccessLimited=true
```

Get connected patients with blocked GP data access:

```
GET /organisations/ORG123/patients?  
connectionStatus=Connected&gpDataAccessLimited=true
```

Get connected patients with patient-granted access:

```
GET /organisations/ORG123/patients?  
connectionStatus=Connected&dataAccessStatus=Reviewed
```

Get patients with date of birth mismatch:

```
GET /organisations/ORG123/patients?connectionStatus=DataMismatch&dobMismatch=true
```

Get patients with surname mismatch:

```
GET /organisations/ORG123/patients?  
connectionStatus=DataMismatch&surnameMismatch=true
```

Get first 100 patients (with pagination):

GET /organisations/ORG123/patients?limit=100&offset=0

Get next 50 patients:

```
GET /organisations/ORG123/patients?limit=50&offset=100
```

Responses

Status	Meaning
200	OK
400	Invalid pagination parameters (limit > 1000, limit < 1, or offset < 0), or invalid filter value for connectionStatus, dataAccessStatus, patientDataAccessLimited, gpDataAccessLimited, dobMismatch, or surnameMismatch, or invalid combination of these parameters (for example, patientDataAccessLimited or gpDataAccessLimited cannot be set if connectionStatus is not Connected)
404	Organisation not found

Update Patient

PATCH /organisations/{extOrganisationId}/patients/{extPatientId}

Updates one or more demographic attributes of an existing patient within an organisation. This is typically used when the EHR updates patient details such as contact information or postcode. The same validation and overwrite rules apply as described in **Data priority and overwrite rules (ORB vs EHR data)** (e.g. date of birth and surname are validated, not overwritten).

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)
extPatientId	External patient ID (EHR)

```
{
  "title": "string",
  "firstName": "string",
  "lastName": "string",
  "dateOfBirth": "string",
  "sex": "string",
  "emailAddress": "string",
  "postcode": "string"
}
```

Request Body

Request Body Fields

Field	Type	Required	Description
title	string	No	Patient title
firstName	string	No	Patient's first name
lastName	string	No	Patient's last name (surname)
dateOfBirth	string	No	Date of birth in YYYY-MM-DD format
sex	string	No	Patient sex (e.g., Male, Female)
emailAddress	string	No	Patient email address
postcode	string	No	Patient postcode

When updating an existing patient:

- ♦ If **dateOfBirth** or **lastName** are supplied, they are validated against existing ORB data. If they do not match, the request fails with a 409 validation error (these fields are not overwritten).
- ♦ Any other supplied fields overwrite ORB data according to the **Data priority and overwrite rules (ORB vs EHR data)** (NHS data remains the source of truth where present).

Response Body (200)

On success, ORB returns the patient's **current demographic fields** as stored after the update (not an empty body). Values may differ from the request where NHS data takes precedence or where **dateOfBirth** / **lastName** were validated but not overwritten.

```
{
  "title": "Ms",
  "sex": "Male",
  "dateOfBirth": "1933-11-14",
  "firstName": "Simon",
  "lastName": "OLLEY",
  "postcode": "asdas123",
  "emailAddress": "patient@example.com"
}
```

Response Body Fields (200)

Field	Type	Description
title	string	Patient title (e.g. Mr, Ms)
sex	string	Patient sex (e.g. Male, Female)
dateOfBirth	string	Date of birth in YYYY-MM-DD format
firstName	string	Patient's first name
lastName	string	Patient's last name (surname)
postcode	string	Patient postcode
emailAddress	string	Patient email address (may be null in ORB if not set)

This response does **not** include **extPatientId**, **connectionStatus**, or other list-only fields (see **List Patients**).

Responses

Status	Meaning
200	Patient updated
400	Update failed due to invalid request or validation error
404	Patient or organisation not found
409	Conflict — request body DOB/surname does not match existing ORB patient record (ORB vs request mismatch).
500	Unexpected server error

Connect Patient by Email

POST /organisations/{extOrganisationId}/patients/{extPatientId}/connection-email

Sends an email inviting a patient to sign up and grant permission for the organisation to view NHS records.

- If the patient does not exist, ORB creates a new patient record and sends a NHS invitation email.
- If the patient exists but is not connected, ORB sends the invitation email directly (unless resend=false and an invitation is already pending). In this case, ORB ensures the patient email, date of birth and surname matches the ones in the request. Remaining fields are updated in ORB, using the data passed in the request.
- If the patient is already connected, it sends a data access request email for the patient to grant organisation access to their NHS (unless access is already granted, or the data access request rate limit for the last 24 hours has already been reached).

The EHR may call this endpoint again to send a **reminder** (e.g. if the patient has not yet completed NHS login or granted access). Whether an email is sent is subject to the rate limits in the Patients section above: invitation can be resent when **resend** is true; data access request emails are sent at most twice per patient per organisation per 24 hours.

This includes the ID of the requesting Clinician User, so ORB can include this in the email.

The invitation opens ORB's patient-facing NHS linking page. Logo and welcome text for that page are configured via Set Organisation Branding.

The NHS registration link in the email expires after the configured invitation period (**7 days by default**). Sending a replacement registration email generates a new link, invalidates the previous link, and restarts the expiry period. A data-access / consent email uses a login URL without a separate link expiry.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)
extPatientId	External patient ID (EHR)

Request Body

```
{
  "title": "string",
  "firstName": "string",
  "lastName": "string",
  "dateOfBirth": "string",
  "sex": "string",
  "emailAddress": "string",
  "postcode": "string",
  "requestingClinicianId": "string",
  "resend": boolean
}
```

Request Body Fields

Field	Type	Required	Description
title	string	Yes	Patient title
firstName	string	Yes	Patient's first name
lastName	string	Yes	Patient's last name
dateOfBirth	string	Yes	Date of birth in YYYY-MM-DD format
sex	string	Yes	Patient sex (e.g., Male, Female)
emailAddress	string	Yes	Patient email address
postcode	string	Yes	Patient postcode
requestingClinicianId	string	Yes	External user ID of the requesting clinician.
resend	boolean	No	True to force resend of the email if already invited

Responses

Status	Meaning
200	Invite sent
201	Patient already invited. No invite sent unless resend flag is true, but data access request is sent, when applies
400	Connection failed due to invalid request or validation error
404	Organisation or requesting clinician not found
409	Conflict — external patient ID already in use by another patient, or request body DOB/surname does not match existing ORB patient record (ORB vs request mismatch). Distinct from 422 (NHS→ORB mismatch, see below).
422	Returned when the patient's connection status is DataMismatch (NHS login revealed a mismatch between ORB and NHS data). Response body includes connectionStatus DataMismatch with dobMismatch/surnameMismatch . Distinct from 409 (request body does not match ORB, or external patient ID already in use).
429	Rate limit exceeded: the maximum number of connection emails allowed per patient per organisation in 24 hours has been reached.

Under discussion: Use of 200 vs 201 for these responses is under review (e.g. consistently returning 200 with the response body indicating whether an email was sent or the patient was created, and reserving 201 for the case where the patient resource was created).

Response Body (200 / 201)

On success, the response body returns the current connection and data access status:

```
{
  "connectionStatus": "string",
  "dobMismatch": false,
  "surnameMismatch": false,
  "patientDataAccessLimited": false,
  "gpDataAccessLimited": false,
  "dataAccessStatus": "string",
  "oldestNhsRecordDate": "string",
  "lastInvitationEmailDate": "string",
  "lastAccessRequestEmailDate": "string"
}
```

Response Body Fields (200 / 201)

Field	Type	Description
connectionStatus	string	Connection status (InviteNotSent, InviteSent, InviteExpired, DataMismatch, Connected)

Field	Type	Description
dobMismatch	boolean	When connectionStatus is DataMismatch: true if date of birth mismatch detected (optional)
surnameMismatch	boolean	When connectionStatus is DataMismatch: true if surname mismatch detected (optional)
patientDataAccessLimited	boolean	When applicable: true if the patient has restricted their data partially or totally (optional)
gpDataAccessLimited	boolean	When applicable: true if the GP has blocked the patient's data partially or totally (optional)
dataAccessStatus	string	Data access request status (RequestNotSent, RequestSent, Reviewed) (optional)
oldestNhsRecordDate	string	Date of oldest NHS record received (ISO-8601); null if none or not determined (optional)
lastInvitationEmailDate	string	ISO-8601 date of the last NHS registration invitation email sent (optional)
lastAccessRequestEmailDate	string	ISO-8601 date of the last data access request email sent (optional)

Error Response Body (422)

When a validation error occurs (422 status), the response body looks the same as the success response, with **connectionStatusDataMismatch** and **dobMismatch**/ **surnameMismatch** indicating which validation failed:

```
{
  "connectionStatus": "DataMismatch",
  "dobMismatch": true,
  "surnameMismatch": false,
  "patientDataAccessLimited": false,
  "gpDataAccessLimited": false,
  "dataAccessStatus": "string",
  "oldestNhsRecordDate": "string",
  "lastInvitationEmailDate": "string",
  "lastAccessRequestEmailDate": "string"
}
```

Error Response Body Fields (422)

Field	Type	Description
connectionStatus	string	DataMismatch (validation error)
dobMismatch	boolean	true if date of birth mismatch detected
surnameMismatch	boolean	true if surname mismatch detected
patientDataAccessLimited	boolean	Optional; when applicable, true if patient has restricted their data
gpDataAccessLimited	boolean	Optional; when applicable, true if GP has blocked patient data
dataAccessStatus	string	Optional; RequestNotSent, RequestSent, or Reviewed
oldestNhsRecordDate	string	Optional; ISO-8601 date
lastInvitationEmailDate	string	Optional; ISO-8601 date
lastAccessRequestEmailDate	string	Optional; ISO-8601 date

Connect Patient by Link

POST /organisations/{extOrganisationId}/patients/{extPatientId}/connection-link

Returns an invitation link, which can be sent to a patient as part of a link request.

- If the patient does not exist, ORB creates a new patient record, a NHS invitation link is built and returned.
- If the patient exists but is not connected, a NHS invitation link is built and returned. In this case, ORB ensures the patient email, date of birth and surname matches the ones in the request. Remaining fields are updated in ORB, using the data passed in the request.
- If the patient is already connected, it is indicated with response code 201 and no link for NHS invitation is returned.

In all previous cases, if a data access request link can still be issued under the rate limits, a data access request link is built and returned too.

Calling this endpoint again allows the EHR to obtain a fresh link to send as a **reminder** to the patient. Invitation and data access request links are only returned while the respective limits in the last 24 hours for this patient and organisation have not been exceeded (see **Rate limits** in the Patients section above).

The returned invitation link opens ORB's patient-facing NHS linking page. Logo and welcome text for that page are configured via Set Organisation Branding.

An NHS registration link expires after the configured invitation period (**7 days by default**). Generating a replacement invalidates the previous registration link and restarts the expiry period. A returned data-access / consent link is an authenticated ORB login URL and has no separate link expiry.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)
extPatientId	External patient ID (EHR)

Request Body

```
{
  "title": "string",
  "firstName": "string",
  "lastName": "string",
  "dateOfBirth": "string",
  "sex": "string",
  "emailAddress": "string",
  "postcode": "string",
  "requestingClinicianId": "string",
  "resend": boolean
}
```

Request Body Fields

Field	Type	Required	Description
title	string	Yes	Patient title
firstName	string	Yes	Patient's first name
lastName	string	Yes	Patient's last name
dateOfBirth	string	Yes	Date of birth in YYYY-MM-DD format
sex	string	Yes	Patient sex (e.g. Male, Female)
emailAddress	string	Yes	Patient email address
postcode	string	Yes	Patient postcode
requestingClinicianId	string	Yes	External user ID of the requesting clinician

Response Body

```
{
  "invitationLink": "string",
  "dataAccessRequestLink": "string",
  "connectionStatus": "string",
  "dobMismatch": false,
  "surnameMismatch": false,
  "patientDataAccessLimited": false,
  "gpDataAccessLimited": false,
  "dataAccessStatus": "string",
  "oldestNhsRecordDate": "string",
  "lastInvitationEmailDate": "string",
  "lastAccessRequestEmailDate": "string"
}
```

Response Body Fields

Field	Type	Description
invitationLink	string	URL to the patient invitation that can be shared or displayed
dataAccessRequestLink	string	URL for the patient to grant data access to the organisation
connectionStatus	string	Connection status (InviteNotSent, InviteSent, InviteExpired,
dobMismatch	boolean	When connectionStatus is DataMismatch: true if date of birth
surnameMismatch	boolean	When connectionStatus is DataMismatch: true if surname
patientDataAccessLimited	boolean	When applicable: true if the patient has restricted their data
gpDataAccessLimited	boolean	When applicable: true if the GP has blocked the patient's data
dataAccessStatus	string	Data access request status (RequestNotSent, RequestSent,
oldestNhsRecordDate	string	Date of oldest NHS record received (ISO-8601); null if none or
lastInvitationEmailDate	string	ISO-8601 date of the last NHS registration invitation email sent
lastAccessRequestEmailDate	string	ISO-8601 date of the last data access request email sent

Responses

Status	Meaning
200	Patient was not invited yet, or invitation expired. Link for NHS invitation and data access returned.
201	Patient already invited. No invite link returned unless resend flag is true, but data access request link is returned, when applies
400	Connection failed due to invalid request or validation error
404	Organisation or requesting clinician not found
409	Conflict — external patient ID already in use by another patient, or request body DOB/surname does not match existing ORB patient record (ORB vs request mismatch). Distinct from 422 (NHS→ORB mismatch, see below).
422	Returned when the patient's connection status is DataMismatch (NHS login revealed a mismatch between ORB and NHS data). Response body includes connectionStatus DataMismatch with dobMismatch/surnameMismatch . Distinct from 409 (request body does not match ORB, or external patient ID already in use).
429	Rate limit exceeded: the maximum number of connection links allowed per patient per organisation in 24 hours has been reached.

Under discussion: Use of 200 vs 201 for these responses is under review (e.g. consistently returning 200 with the response body indicating whether a link was returned or the patient was created, and reserving 201 for the case where the patient resource was created).

Error Response Body (422)

When a validation error occurs (422 status), the response body looks the same as the success response, with **connectionStatus** **DataMismatch** and **dobMismatch**/**surnameMismatch** indicating which validation failed:

```
{
  "invitationLink": "string",
  "dataAccessRequestLink": "string",
  "connectionStatus": "DataMismatch",
  "dobMismatch": true,
  "surnameMismatch": false,
  "patientDataAccessLimited": false,
  "gpDataAccessLimited": false,
  "dataAccessStatus": "string",
  "oldestNhsRecordDate": "string",
  "lastInvitationEmailDate": "string",
  "lastAccessRequestEmailDate": "string"
}
```

Error Response Body Fields (422)

Field	Type	Description
invitationLink	string	May be null or empty
dataAccessRequestLink	string	May be null or empty
connectionStatus	string	DataMismatch (validation error)
dobMismatch	boolean	true if date of birth mismatch detected
surnameMismatch	boolean	true if surname mismatch detected
patientDataAccessLimited	boolean	Optional; when applicable, true if patient has restricted their data
gpDataAccessLimited	boolean	Optional; when applicable, true if GP has blocked patient data
dataAccessStatus	string	Optional; RequestNotSent, RequestSent, or Reviewed
oldestNhsRecordDate	string	Optional; ISO-8601 date
lastInvitationEmailDate	string	Optional; ISO-8601 date
lastAccessRequestEmailDate	string	Optional; ISO-8601 date

In case of any other error, a body with a single **errorMessage** property is returned, describing the error in a human-readable format.

Get Patient Connection Status

GET /organisations/{extOrganisationId}/patients/{extPatientId}/connection

Returns the connection status and access levels for the patient. The response body uses the shape below.

The response may include `nhsNumber`: the patient's NHS number from ORB's NHS Login record (verified via NHS Login). This field is read-only for the EHR — it is not supplied on connect or update requests. `nhsNumber` is omitted or null when the patient has no NHS Login on record.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)
extPatientId	External patient ID (EHR)

Response Body

```
{
  "connectionStatus": "string",
  "dobMismatch": boolean,
  "surnameMismatch": boolean,
  "patientDataAccessLimited": boolean,
  "gpDataAccessLimited": boolean,
  "dataAccessStatus": "string",
  "nhsNumber": "string"
  "oldestNhsRecordDate": "string",
  "lastInvitationEmailDate": "string",
  "lastAccessRequestEmailDate": "string"
}
```

When `connectionStatus` is **DataMismatch**, `dobMismatch` and `surnameMismatch` indicate which validation failed.

Response Body Fields

Field	Type	Description
connectionStatus	string	Connection status (see Possible Status Values below)
dobMismatch	boolean	When connectionStatus is DataMismatch: true if date of birth
surnameMismatch	boolean	When connectionStatus is DataMismatch: true if surname
patientDataAccessLimited	boolean	When applicable: true if the patient has restricted their data
gpDataAccessLimited	boolean	When applicable: true if the GP has blocked the patient's data
dataAccessStatus	string	Data access request status (RequestNotSent, RequestSent,
nhsNumber	string	NHS number from NHS Login in ORB (read-only; optional; omitted or null if not on record)
oldestNhsRecordDate	string	Date of oldest NHS record received (ISO-8601); null if none or
lastInvitationEmailDate	string	ISO-8601 date of the last NHS registration invitation email sent
lastAccessRequestEmailDate	string	ISO-8601 date of the last data access request email sent

Possible Status Values (connectionStatus)

Status	Description
InviteNotSent	Invitation for the patient to log in using NHS credentials is not sent yet
InviteSent	Invite has been sent to the patient
InviteExpired	Invitation has been sent but expired
DataMismatch	Validation error (date of birth and/or surname mismatch); see dobMismatch and surnameMismatch
Connected	Patient is connected to the organisation

Responses

Status	Meaning
200	OK
400	Bad request
404	Patient or organisation not found

Get Granted Permissions

GET /organisations/{extOrganisationId}/patients/{extPatientId}/permissions

Returns the list of data permissions that the patient has granted to this organisation. This endpoint requires that the patient is connected to the organisation (see Get Patient Connection Status).

Connection Status Requirements:

- If the patient connection status is **Connected**, this endpoint returns the granted permissions
- If the connection status is **InviteSent** or **DataMismatch**, this endpoint returns **404 Not Found**(no connection established yet)

This can be used by the EHR to determine what data will be available before requesting an access token, or to display permission status to clinicians.

Note: The access token request endpoint will automatically validate permissions before issuing a token. This endpoint is informational only.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)
extPatientId	External patient ID (EHR)

Response Body

```
[
  "Allergies",
  "Medication History",
  "Immunisations",
  "Problems",
  "Documents",
  "Consultation History",
  "TestResults"
]
```

Response Body Fields

Field	Type	Description
[]	array of string	Array of permission names granted to the patient

Possible Permission Values

- Allergies
- Medication History
- Immunisations
- Problems
- TestResults
- Documents
- Consultation History
-

Responses

Status	Meaning
200	OK - Returns array of permissions
400	Bad request
403	Organisation not authorised to view this patient
404	Patient or organisation not found, or patient connection not established (InviteSent, DataMismatch)

Remove Patient Connection

DELETE /organisations/{extOrganisationId}/patients/{extPatientId}/connection

Removes the link between the patient and the organisation. Use this when a patient leaves a clinic; doing so is treated as an implied retraction of the organisation's permission to view that patient's records. Whether to call this API is for the EHR or organisation to decide.

Behaviour: This is an unlink only for the organisation in the request:

- The EHR connection for this organisation is removed, so the patient no longer appears in List Patients for that organisation, and token/connection/status/permissions calls for this patient and organisation fail.
- The patient-to-organisation link for that organisation is also removed, so the patient can no longer access that organisation.
- The underlying patient account, PatientLogin, and NHS/IM1 data are retained. Other organisation links for the same patient are untouched.
- If the patient still has one or more other organisations, they can continue to log in and use those organisations.
- If this was the patient's last organisation, they cannot log in until they are linked to at least one organisation again. The patient account and any completed NHS registration are kept (not deleted).
- If the same patient is connected again later, ORB treats them as an existing patient (matched by email, with DOB/surname validation). Because NHS login already exists, re-connect returns a data-access / data-share link rather than a new NHS invitation link.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)
extPatientId	External patient ID (EHR)

Responses

Status	Meaning
200	Connection removed
400	Deletion failed due to invalid request or validation error
404	Patient or organisation not found
500	Unexpected server error

A successful response (200) returns no response body.

Patient Access

The Patient Access API enables secure viewing of NHS patient records within an EHR system via an embedded iFrame widget. Issuing a patient access token requires the organisation contract to be signed ("403"

OrganisationContractNotSigned otherwise). The security model ensures that:

1. **Server-side token generation:** Tokens are only issued to authenticated EHR backend systems, never directly to browsers
2. **User identification:** Each token request identifies the specific clinician requesting access
3. **Authorisation checks:** ORB validates that the user, organisation, and patient relationships are valid and authorised
4. **Short-lived tokens:** Tokens expire in 15 minutes to limit exposure
5. **Scoped access:** Tokens are bound to specific user-patient-organisation combinations
6. **Audit trail:** All token requests are logged for compliance and security monitoring

Typical Flow:

1. Clinician clicks "View NHS Records" button in EHR patient record page
2. EHR backend makes server-side API call to request access token (including clinician's extUserId)
3. ORB validates authorisation and returns JWT token
4. EHR backend passes token to frontend JavaScript widget
5. Widget loads in iframe and authenticates using the token
6. If the clinician has not yet accepted Terms and Conditions (T&C) for this organisation, the widget shows the T&C screen; after acceptance, NHS data can be viewed
7. Widget displays patient data. Token expires after 15 minutes, requiring a new access-token request for continued access

Widget Integration:

The ORB widget is a JavaScript component that must be embedded in the EHR's patient record page. Embedding, token handling, and runtime behaviour are documented in the [ORB Widget Integration Guide](#). The token is passed to the widget via `postMessage`(recommended; see that guide). Do not call `/access-token` from the browser.

Clinician Terms and Conditions (widget)

Clinician widget Terms and Conditions are required before NHS data is shown in the widget. See [Terms and Conditions]([#terms-and-conditions](#)).

Important Security Considerations:

- ◆ Tokens must never be logged, stored in browser storage, or exposed in URLs visible to users
- ◆ The EHR backend should validate that the requesting user has permission to view the patient before requesting a token
- ◆ If a token expires while a user is viewing data, the widget will handle re-authentication automatically Suspended
- ◆ organisations (and therefore suspended users) cannot request access tokens

Request Access Token

POST /organisations/{extOrganisationId}/patients/{extPatientId}/access-token

Returns a short-lived JWT used to access patient data via the ORB widget (iframe).

Security Model:

- The EHR must make this request server-side (never from the browser) when a clinician clicks to view patient data
- The request must include the `extUserId` of the clinician requesting access ORB
- validates that:
 - The user exists and belongs to the specified organisation
 - The user has `**`isClinician: true`**` (non-clinical users cannot request access tokens)
 - The organisation is not suspended
 - The organisation has access to the patient
 - The patient has granted the necessary permissions
- The returned JWT token is scoped to the specific user, patient, and organisation
- The token has a short expiration time (15 minutes) to limit exposure if intercepted
- The token should be passed to the widget immediately and not stored or logged

Issuing an access token does **not** mean the clinician has accepted widget T&C. Token issuance only establishes that the EHR backend is allowed to open the widget for that user, patient, and organisation. T&C are enforced when the widget loads and before NHS data is returned from widget APIs — see [Terms and Conditions](#terms-and-conditions).

Requesting an access token is recorded as clinician access to that patient for the organisation. ORB uses this (together with patient access) for data access review and retention policies.

Path Parameters

Name	Description
extOrganisationId	External organisation ID (EHR)
extPatientId	External patient ID (EHR)

Request Body

```
{
  "extUserId": "string"
}
```

Request Body Fields

Field	Type	Required	Description
extUserId	string	Yes	External user ID (EHR) of the requesting user

Response Body

```
{
  "accessToken": "string",
  "expiresIn": 900
}
```

Response Body Fields

Field	Type	Description
accessToken	string	Short-lived JWT token for accessing patient data (expires in expiresIn seconds)
expiresIn	number	Token expiration time in seconds (typically 900 = 15 minutes)

JWT Token Claims

The JWT token contains the following claims:

- ♦ **extOrganisationId**- The organisation ID
- ♦ **extUserId**- The clinician user ID who requested access
- ♦ **extPatientId**- The patient ID
- ♦ **iat**- Issued at timestamp
- ♦ **exp**- Expiration timestamp
- ♦ **aud**- Audience (e.g., "orb-widget")
- ♦ **iss**- Issuer (e.g., "orb-api")

Responses

Status	Meaning
200	Token issued successfully
400	Invalid request (missing extUserId)
403	Access denied — user not authorised (including `isClinician: false`), organisation suspended, or patient permissions insufficient (`errorCode`: `NonClinicianAccess` when the user is non-clinical)
404	Patient, user, or organisation not found
500	Internal server error (see Common Errors)

Terms and Conditions

ORB requires agreement acceptance in more than one place. This section describes those agreements. They are separate processes and must not be confused.

Agreement	Who accepts	Status in this guide
Clinician widget T&C	Each clinician user, per organisation	Documented below
Organisation contract (clinic)	Authorised organisation representative (host-controlled)	Documented below

Clinician Terms and Conditions (widget)

Before NHS patient data is shown in the embedded widget, each clinician user must accept ORB Clinician Terms and Conditions (T&C) for the organisation they are acting in.

This is the clinician widget T&C only (tick-box acceptance in the iframe). It is not the organisation-level clinic contract.

Agreement type and scope

Item	Behaviour
Mechanism	Simplified tick-box (checkbox) acceptance inside the ORB-hosted widget UI
Scope	Acceptance is recorded per clinician user per organisation
Multi-org clinicians	If the same clinician works in multiple organisations in the EHR, they must accept T&C separately for each ORB organisation
T&C content	System-defined wording hosted on ORB as an editable Static Page (alias widget-terms-and-conditions), so ORB can update the text without an EHR release
Patients	Clinician widget T&C does not apply to patients

Trigger and mandatory acceptance

- The agreement is presented when the clinician first opens / uses the widget for that organisation (after a valid access token has been passed into the iframe).
- Acceptance is mandatory: the clinician cannot reject the terms and continue. NHS record data and related widget data APIs remain blocked until acceptance is completed.
- Re-acceptance: once accepted for a given user–organisation pair, the clinician is not required to re-accept T&Cs for the same organisation (acceptance is idempotent).

What ORB records

On acceptance, ORB records that the clinician approved the T&Cs for that user–organisation pair (including when it was accepted), for audit and compliance. Until acceptance is completed, widget IM1/data reads

return 403 with errorCode **TermsNotAccepted**.

Hosting and context

- T&C pages / content are hosted by ORB (MiPro / ORB sites) and shown inside the third-party EHR iframe (e.g. Semble).
- Clinician and organisation context for acceptance comes from the widget access token (scoped user, organisation, patient), not from calling External API T&C endpoints.
- PDF download of the current T&C wording is also served from ORB for the widget session (ORB-internal widget APIs).

Enforcement vs access token

Step	Clinician T&C status
EHR calls POST .../access-token	Clinician T&C not checked (organisation contract is required — see Organisation contract)
Widget loads and clinician uses medical-record views	Clinician T&C enforced; UI shows T&C screen if not yet accepted; data APIs blocked until accept

EHR integrators: Acceptance and enforcement are handled inside the ORB widget after the EHR passes the access token. The ORB External API does not expose endpoints to accept clinician T&C or to query acceptance status.

Organisation contract (clinic)

- As part of iFrame-related onboarding, each EHR organisation must complete a separate organisation-level ORB contract / Terms & Conditions signing flow before protected API and Widget functionality for that organisation is available.
- The signing UI is hosted by ORB and embedded in an iframe controlled by the host EHR (e.g. Semble). Contract content, signing logic, audit logging, PDF generation, and download links are handled on the ORB side.
- The host controls when and for which organisation the contract iframe is shown, and which user sees the launch button.
- The host should show the contract flow only to the intended authorised organisation representative.
- Name and email submitted in the iframe are recorded as supplied (they may be pre-filled via SET_SIGNATORY and edited by the signatory).

Scope

Item	Behaviour
Scope	Acceptance is recorded per organisation
Multi-org users	If the same person is linked to multiple organisations, each organisation must sign separately
Content	ORB Static Page (alias organisation-contract-terms-and-conditions), shown inside the iframe
Re-signing	One signature per organisation; a further accept attempt returns 409

Host flow (summary)

- Create the organisation (POST /organisations/{extOrganisationId}) if it does not exist yet.
- Optionally poll GET /organisations/{extOrganisationId}/contract-status.
- When ready, mint POST /organisations/{extOrganisationId}/contract-access-token.
- Open the ORB contract page in an iframe (path /medical-record-embedded/organisation-contract/sign) and pass the token via postMessage SET_TOKEN and send SET_SIGNATORY with { first_name, last_name, email } to pre-fill the form.
- Signatory reviews the contract, enters full name and email, ticks acceptance, and submits.
- On success, the iframe posts CONTRACT_SIGNED with { signedContractUrl }, shows confirmation, and a Download Signed Contract button (; opens the signed PDF via a login-free shared link). A signed PDF is also emailed to the address submitted.
- Host may poll contract-status again or listen for CONTRACT_SIGNED / orb-widget-contract-signed.

After signing

When the contract is signed, ORB stores for that organisation: signatory first and last name (from fullName), email, signing timestamp, IP address, browser / user agent, signed status, and a reference to the signed PDF (stored in object storage; available via a login-free shared download link that does not expire in normal operation). The signed PDF is emailed to the signatory using template code SignedOrbAgreement.

Allowed before the contract is signed

Method	Path	Notes
POST	/organisations/{extOrganisationId}	Create organisation (onboarding)
GET	/organisations	List organisations
GET	/organisations/{extOrganisationId}	Get organisation
GET	/organisations/{extOrganisationId}/contract-status	Poll signing status
POST	/organisations/{extOrganisationId}/contract-access-token	Mint iframe token

Blocked until the contract is signed

Until `contractSigned` is true, protected External API and widget operations for that organisation return 403 with `errorCode OrganisationContractNotSigned` (see Common Errors). Enforcement uses a central action filter on the EHR API (and the widget auth filter for NHS data reads).

Method	Path
PATCH	<code>/organisations/{extOrganisationId}</code>
DELETE	<code>/organisations/{extOrganisationId}</code>
POST	<code>/organisations/{extOrganisationId}/branding</code>
GET	<code>/organisations/{extOrganisationId}/users</code>
POST	<code>/organisations/{extOrganisationId}/users/{extUserId}</code>
PATCH	<code>/organisations/{extOrganisationId}/users/{extUserId}</code>
DELETE	<code>/organisations/{extOrganisationId}/users/{extUserId}</code>
GET	<code>/organisations/{extOrganisationId}/patients</code>
PATCH	<code>/organisations/{extOrganisationId}/patients/{extPatientId}</code>
POST	<code>/organisations/{extOrganisationId}/patients/{extPatientId}/connection-email</code>
POST	<code>/organisations/{extOrganisationId}/patients/{extPatientId}/connection-link</code>
GET	<code>/organisations/{extOrganisationId}/patients/{extPatientId}/connection</code>
DELETE	<code>/organisations/{extOrganisationId}/patients/{extPatientId}/connection</code>
GET	<code>/organisations/{extOrganisationId}/patients/{extPatientId}/permissions</code>
POST	<code>/organisations/{extOrganisationId}/patients/{extPatientId}/access-token</code>
—	Widget NHS patient-data APIs (after a patient access token is used in the iframe)

There is no separate External API “create patient” endpoint; patient onboarding uses the connection and related patient endpoints above, which are blocked until the contract is signed.

Support and API access

- ♦ **API keys and base URLs:** Provided when your integration is onboarded. Use the appropriate base URL for your environment (UAT, Test, or Production).
 - ♦ **Technical support and issues:** For integration support, reporting defects, or questions about this API, contact your ORB integration contact or the support channel provided with your API key.
 - ♦ **Rate limits:** Any rate limits or throttling that apply will be communicated with your API key or in separate operational guidance.
-